

AI Agent in LangChain

This topic brings together everything you've learned so far:

- LLMs
- Prompts
- Chains
- Runnable
- Tools
- Tool Calling
- RAG

Now we combine them to build something much more powerful:

AI Agents

1. Intro

Up to now, most LangChain applications followed predefined workflows.

Example:

User → Prompt → LLM → Output

Or:

User → Retriever → LLM → Output

These workflows are fixed.

But what if we want a system that can:

- Think about a problem
- Decide what to do
- Choose tools
- Use multiple tools
- Change its strategy if needed

That is where AI Agents come in.

2. AI Agent Demo

The demo usually shows an agent handling a complex query like:

"Find the current population of Pakistan and calculate 5% of it."

A normal LLM might:

- Guess the population
- Perform inaccurate calculations

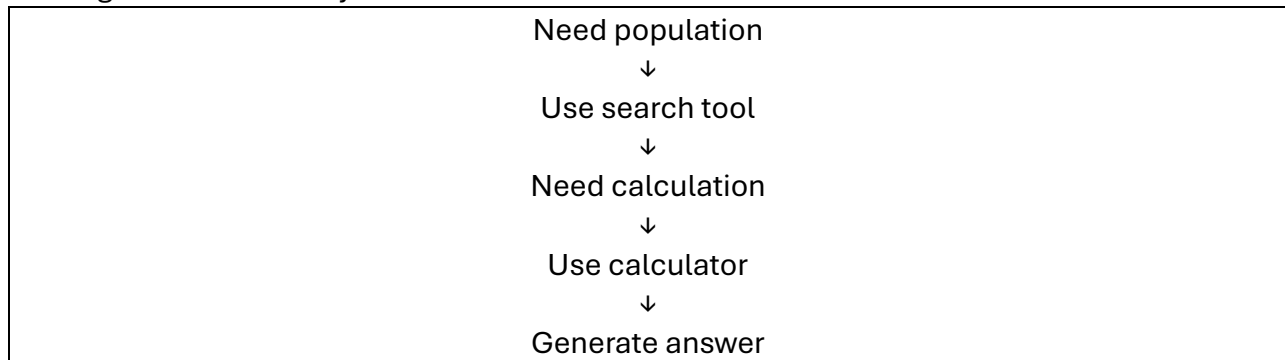
An AI Agent can:

1. Search for current population
 2. Retrieve latest data
 3. Use calculator
 4. Compute 5%
 5. Return accurate answer
-

What Makes This Impressive?

The user gave only one instruction.

The agent automatically decided:



This autonomous decision-making is the essence of an AI Agent.

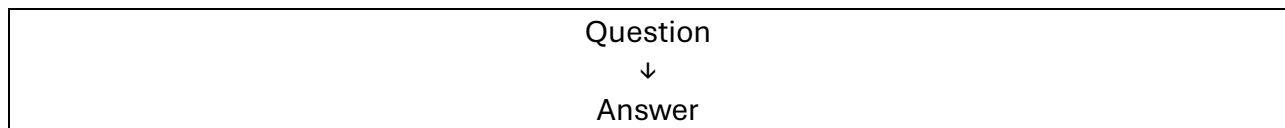
3. What is an AI Agent?

Simple Definition

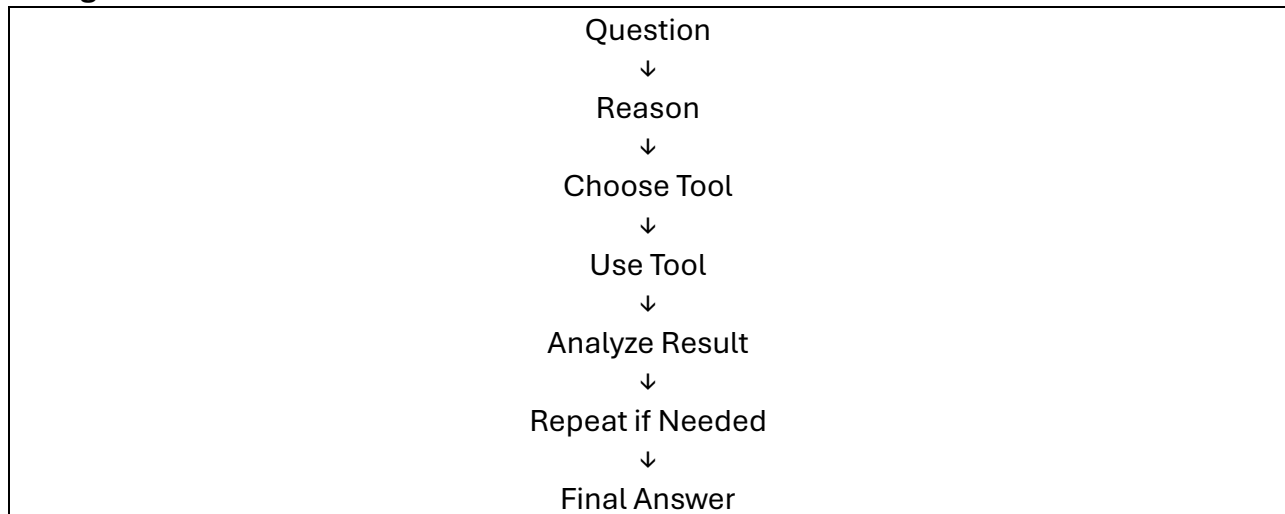
An AI Agent is:

An LLM-powered system that can reason, decide actions, use tools, and work toward achieving a goal.

Traditional LLM



AI Agent



Why Agents Were Needed

Imagine asking:

"What is the weather in Abbottabad and should I carry an umbrella?"

To answer properly:

1. Get weather data

2. Analyze rain probability
3. Make recommendation

A simple prompt may not be enough.

An agent can perform multiple steps automatically.

4. Characteristics of AI Agents

An AI Agent usually has the following characteristics:

1. Reasoning

The agent thinks about the problem.

Example:

Need current weather.

Need weather tool.

2. Planning

The agent creates a plan.

Example:

Step 1: Get weather

Step 2: Analyze

Step 3: Respond

3. Tool Usage

The agent can use external tools.

Examples:

- Search
 - Calculator
 - Database
 - APIs
-

4. Autonomy

The agent decides actions without the developer hardcoding every step.

5. Iteration

The agent can perform multiple rounds of thinking and acting.

Example:

Search

↓

Not enough information

↓

Search again

↓

Answer

5. Coding the AI Agent

An AI Agent typically requires:

1. LLM

The brain.

Example:

```
llm = ChatOpenAI()
```

2. Tools

The abilities.

Example:

```
search_tool  
calculator_tool
```

3. Prompt

Instructions for behavior.

Example:

You are a helpful assistant.

Use tools whenever needed.

4. Agent

The decision-maker.

5. Agent Executor

The runtime engine.

Architecture

```
graph TD
    User --> Agent1[Agent]
    Agent1 --> Tools
    Tools --> Agent2[Agent]
    Agent2 --> Response
```

6. What is ReAct?

This is one of the most important concepts in AI Agents.

ReAct = Reason + Act

The idea was introduced in a research paper.

The agent alternates between:

Reasoning

What should I do?

and
Acting
Use a tool.

ReAct Cycle

Thought
↓
Action
↓
Observation
↓
Thought
↓
Action
↓
Observation
↓
Final Answer

Example

Question:
"What is 15% of Pakistan's population?"

Thought

I need Pakistan's population.

Action

Use search tool.

Observation

Population = 250 million

Thought

Need 15% calculation.

Action

Use calculator.

Observation

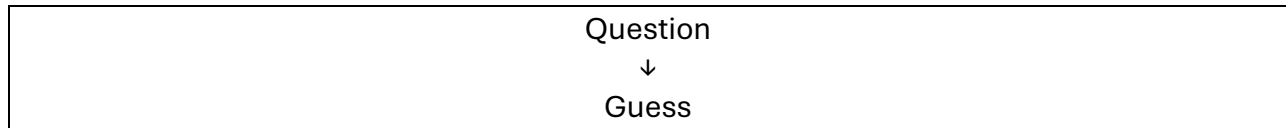
37.5 million

Final Answer

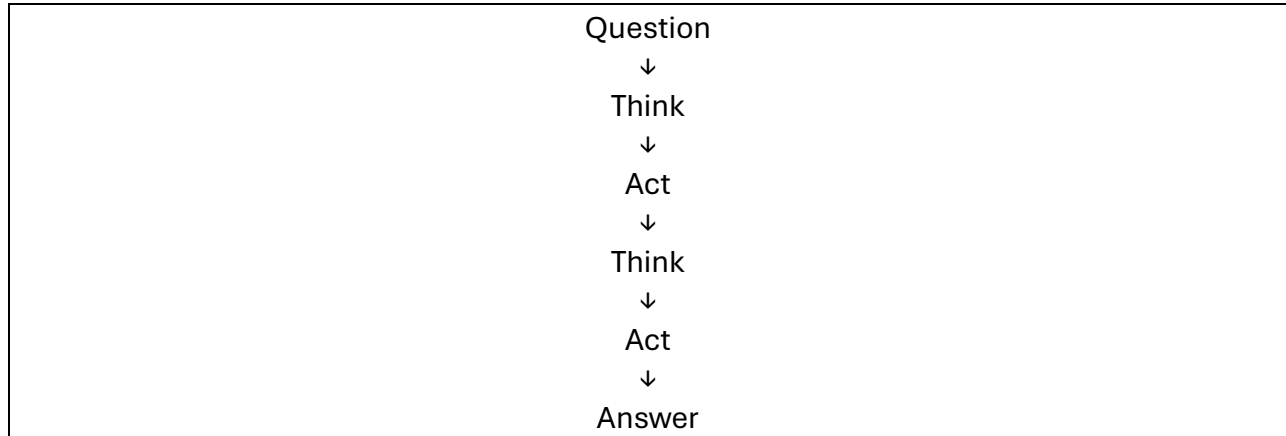
15% of Pakistan's population is approximately 37.5 million.

Why ReAct Matters

Without ReAct:



With ReAct:



7. Agent & Agent Executor

Many beginners confuse these.

Agent

The decision-maker.

Responsible for:

- Reasoning
- Choosing tools
- Planning actions

Think of Agent as:

The Brain

Agent Executor

Responsible for:

- Running the agent
- Executing tools
- Managing workflow

Think of Agent Executor as:

The Manager

Analogy

Imagine a company.

Agent

CEO deciding:
What should we do?

Agent Executor

Operations team doing:
Carry out the plan.

8. Creating an Agent

In LangChain, you create an agent by providing:

LLM

llm

Tools

tools

Prompt

prompt

The agent learns:

Available tools

How to reason

How to act

9. Creating an Agent Executor

After creating the agent:

`agent_executor = AgentExecutor(...)`

The executor handles:

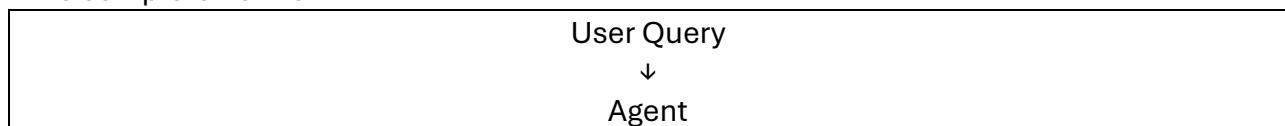
- Tool execution
- Agent iterations
- Intermediate steps
- Final output

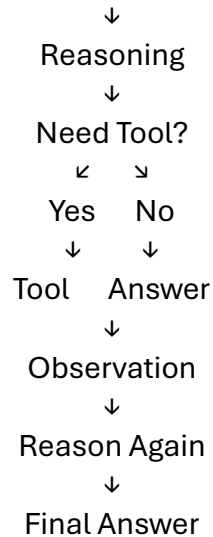
Without Agent Executor:

Agent cannot operate fully.

10. Flow Chart

The complete flow is:





11. Improving the Code

As agents become larger, we improve:

Better Prompts

Clearly define behavior.

Example:

Always verify calculations using tools.

Better Tools

Well-defined descriptions.

Example:

Useful for currency conversion.

Error Handling

Agents should handle:

- API failures
- Missing data
- Invalid inputs

Logging

Useful for debugging.

See:

Thought

Action

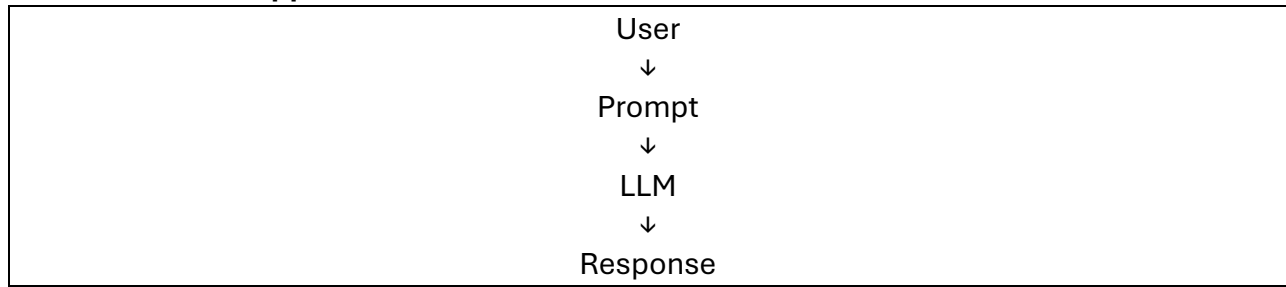
Observation

steps.

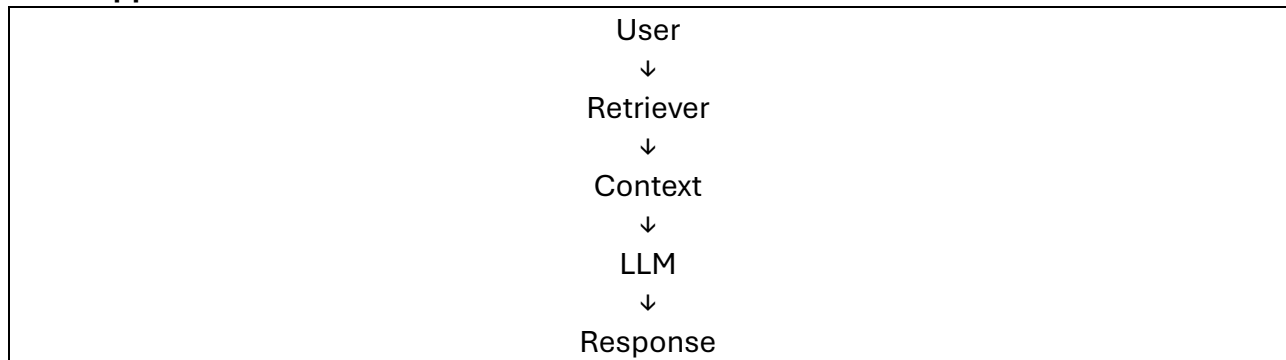
Putting Everything Together

Let's connect all LangChain concepts you've learned.

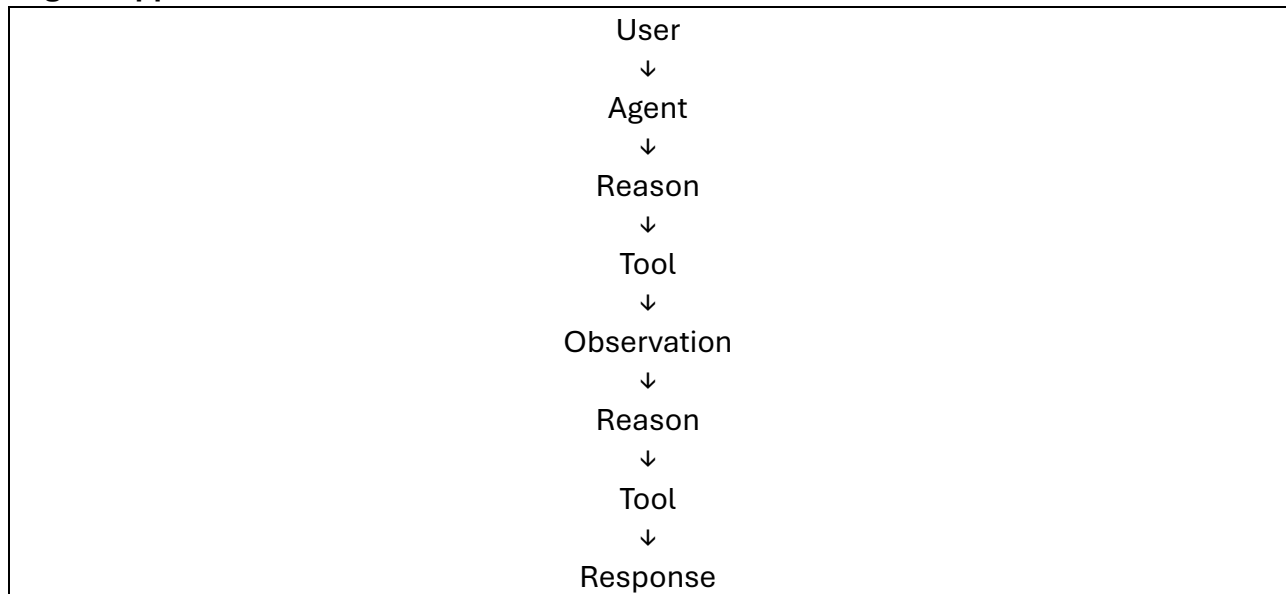
Traditional LLM App



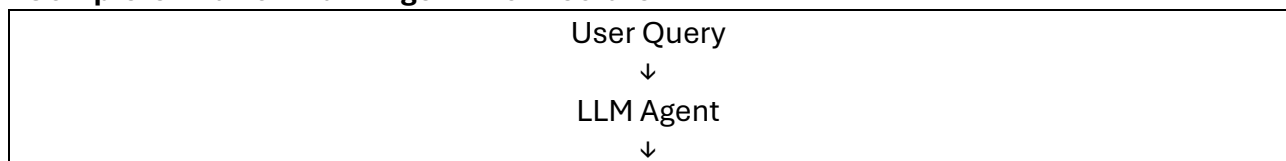
RAG App

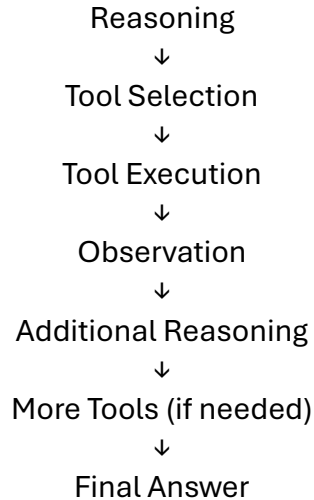


Agent App



Complete End-to-End AI Agent Architecture





Agent vs Chain vs RAG

Feature	Chain	RAG	Agent
Fixed Workflow	✓	✓	✗
Retrieval	✗	✓	Can
Tool Usage	Limited	Limited	✓
Reasoning	Limited	Limited	✓
Planning	✗	✗	✓
Multiple Actions	✗	✗	✓
Autonomous Decisions	✗	✗	✓

Big Picture Understanding

A simple LLM is like:

A student answering from memory.

A RAG system is like:

A student using a textbook.

An AI Agent is like:

A professional worker who can:

- Think
- Search
- Calculate
- Use software
- Make decisions
- Complete tasks

Summary

AI Agent

An intelligent system that can reason, plan, use tools, and take actions.

ReAct

A framework where the agent alternates between:

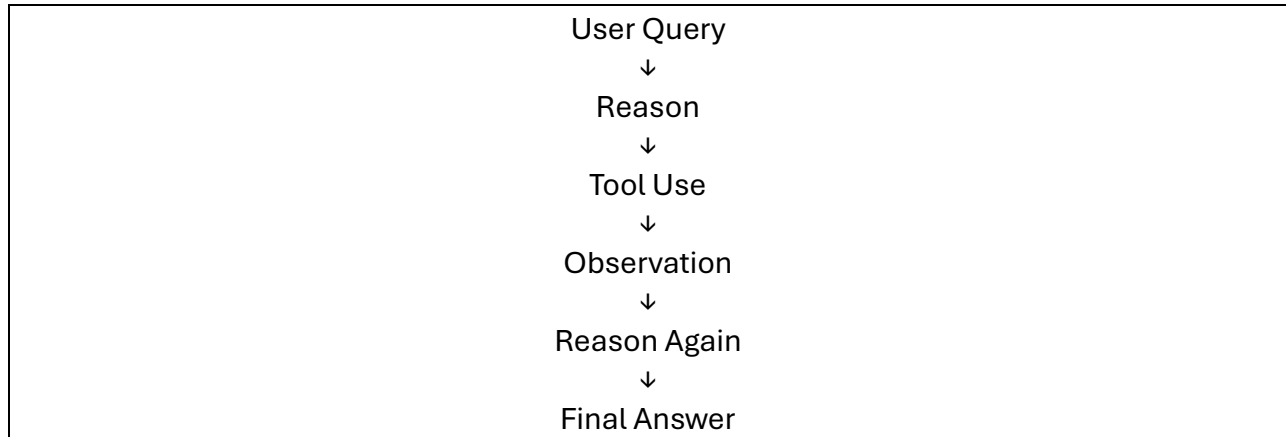
- Thinking (Reason)
- Acting (Tool Use)

Agent

The decision-maker.

Agent Executor

The component that executes actions and manages the workflow.

End-to-End Flow

An AI Agent is an LLM-powered system that can think about a problem, decide what actions to take, use tools to gather information, and continue reasoning until it reaches a goal.